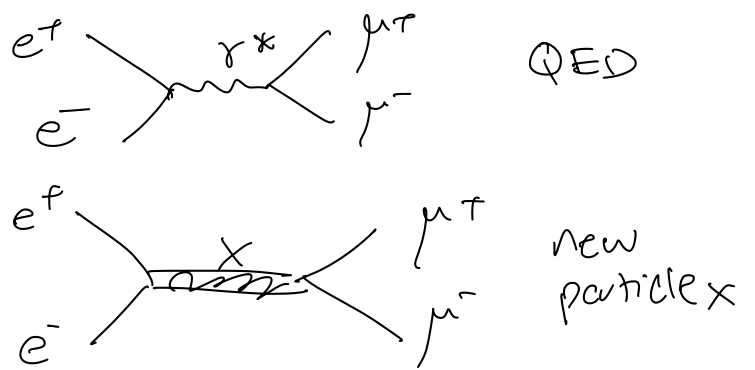
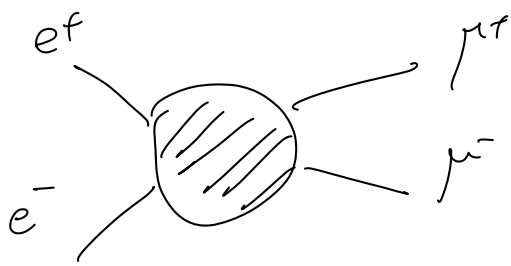


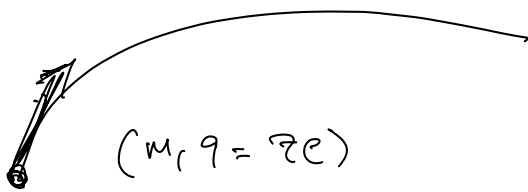


$$m_{inv}^2 = (\underline{p}_{\mu^+} + \underline{p}_{\mu^-})^2$$

if $\mu^-\mu^+$ come from x : $\underline{p}_{\mu^+} + \underline{p}_{\mu^-} = \underline{p}_x$

$|\underline{p}_x|^2 = m_x^2$ in all reference frames.

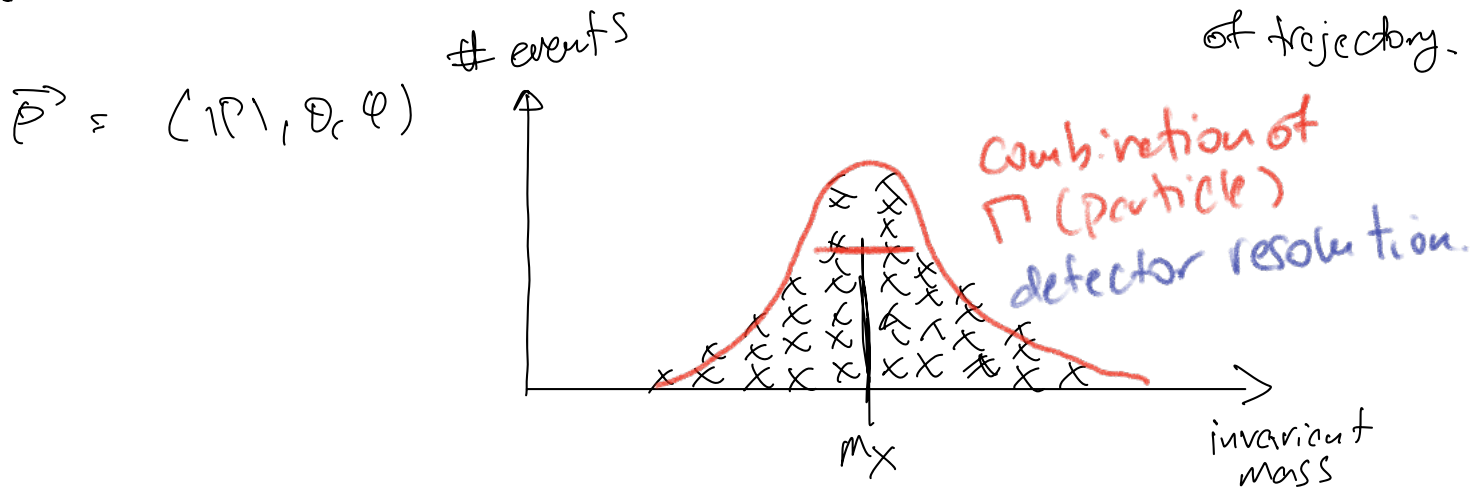
Requires measurement of $E_{\mu^+}, E_{\mu^-}, \vec{p}_{\mu^+}, \vec{p}_{\mu^-}$



$$|\vec{p}| = [0.3 \text{ Z}] B[\text{Tesla}] R[m]$$

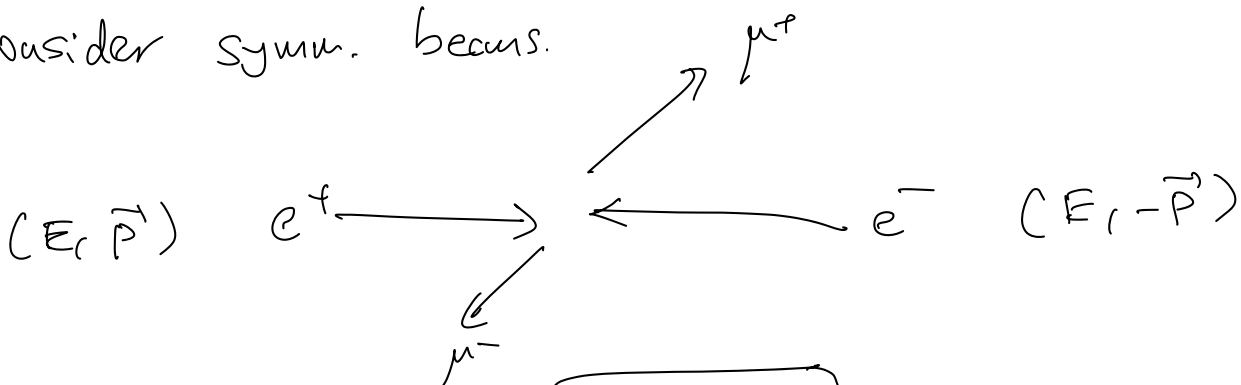
$\downarrow$
 magnetic field.

$\downarrow$
 Curvature or radius of trajectory.



$$\vec{p} = (|\vec{p}|, \theta, \phi)$$

consider symm. beams.



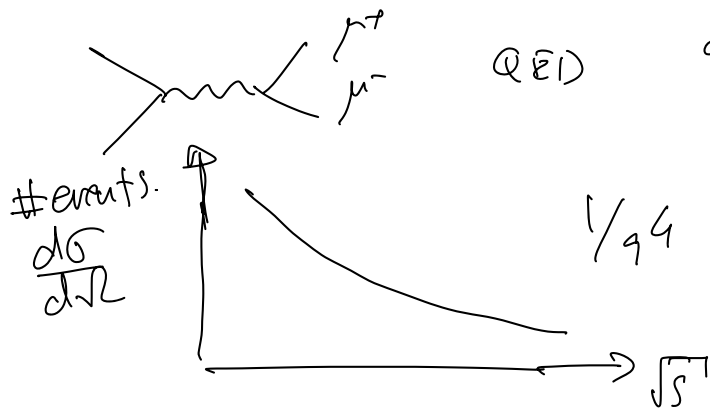
$$\sqrt{s} = \sqrt{(\underline{p}_1 + \underline{p}_2)^2} = \sqrt{(\underline{p}_{\mu^+} + \underline{p}_{\mu^-})^2}$$

initial state $\sqrt{s} = \left((E+E, \vec{0})^2 \right)^{1/2} = 2E$

LHC: $p=E=6.5 \text{ TeV} \Rightarrow \sqrt{s} = 13 \text{ TeV}$

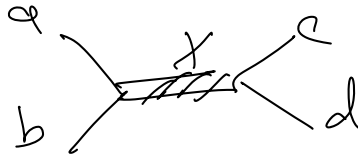
$\sqrt{s}$ energy available to produce x .

$$m_x \leq \sqrt{s}$$



$$\frac{d\sigma}{d\Omega} \sim \underbrace{\left(\frac{\alpha Z_1 Z_2}{q^4} \right)^2}_{\text{Rutherford}} \underbrace{\frac{E'}{E}}_{\text{Recol}} E'^2 \underbrace{(1 - \beta^2 \sin^2 \frac{\theta}{2})}_{\text{Spin of prob. } (e^+e^-)}$$

If you produce x :



$$\sigma(e^+e^- \rightarrow x \rightarrow \mu^+\mu^-) = \frac{2J+1}{(2S_1+1)(2S_2+1)} \frac{4\pi}{|\vec{P}_{in}|^2} \underbrace{\frac{\Gamma^2/4}{(E-m_x)^2 + \Gamma^2/4}}_{\text{Breit-Wigner}} \frac{\Gamma_{in}}{\Gamma_{tot}} \frac{\Gamma_{out}}{\Gamma_{tot}}$$

J : spin of x

s_1 : spin of particle a./ e^-

s_2 : spin of particle b./ e^+

$$|\vec{P}_{in}| = |\vec{P}_{e^+}| = |\vec{P}_{e^-}|$$

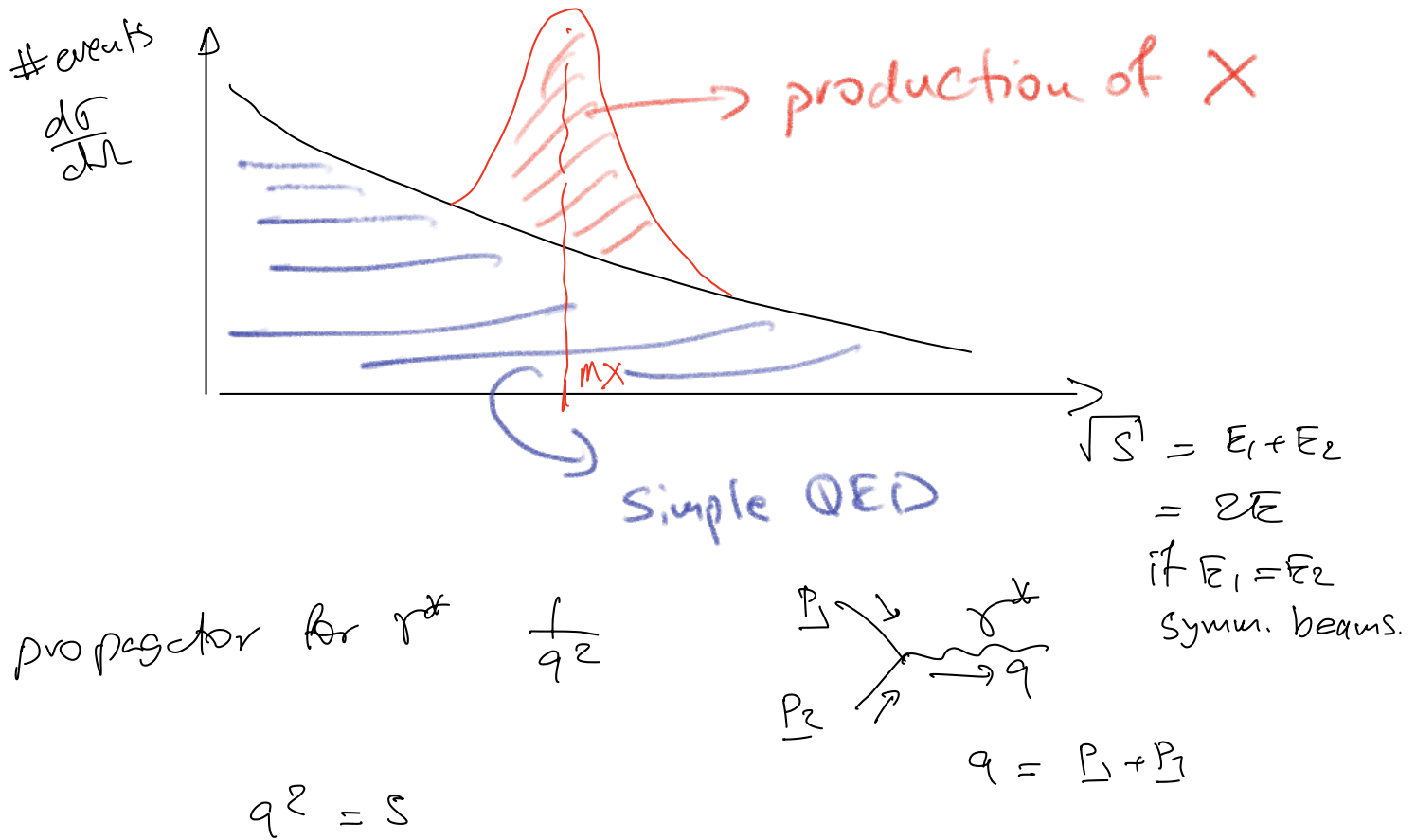
} initial state particles

$$\frac{\Gamma_{in}}{\Gamma_{tot}} = \frac{\Gamma(x \rightarrow e^+e^-)}{\Gamma_{tot}(x)}$$

$$\frac{\Gamma_{out}}{\Gamma_{tot}} = \frac{\Gamma(x \rightarrow \mu^+\mu^-)}{\Gamma_{tot}(x)}$$

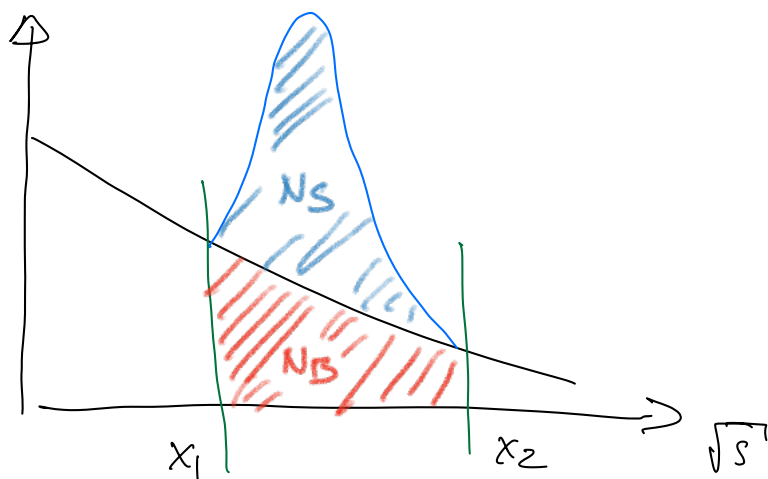
$$\Gamma_{tot} = \sum_i \Gamma_i$$

$$\left. \begin{array}{l} x \rightarrow e^+e^- \\ \rightarrow \mu^+\mu^- \\ \rightarrow \tau^+\tau^- \end{array} \right\} \Gamma = \sum_i \Gamma_i$$



$$\Delta E \cdot \Delta t \approx 1 \Rightarrow \Delta t = \frac{1}{\Delta E}$$

$\Delta E \propto$ invariant mass of virtual photon



$$e^+e^- \rightarrow \mu^+\mu^-$$

$$\frac{dN}{dt} = \sigma \frac{dN_p}{dt} n_{b-d}$$

$$= \sigma \Phi_p N_B$$

$$\sigma = \sigma_{sig} + \sigma_{bg}$$

Signal:

Background:

$$N_{obs} = N_S + N_B$$

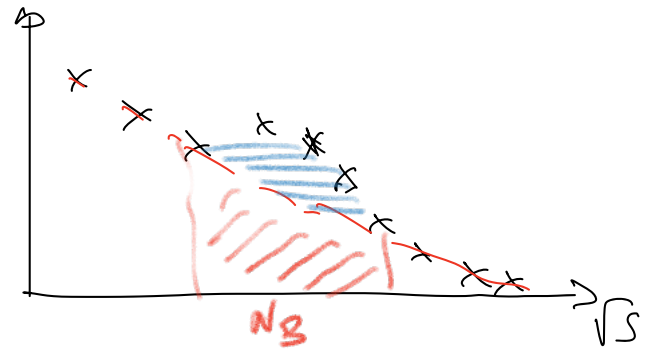
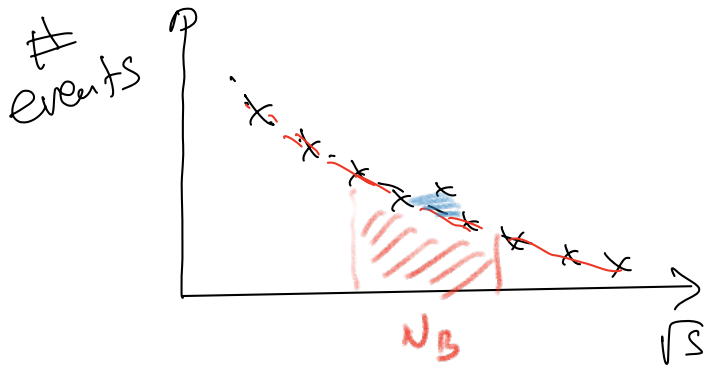
$$\# \text{ events with } \sqrt{s} \in [x_1, x_2]$$

Expected value for N_B

$$\hat{N}_S = N_{obs} - N_B \rightarrow \text{estimated: experimentally theoretical}$$

Counted experimentally.

best estimator for # signal events.



significance of the signal:

N_B : expected background

Poisson variance: $\sqrt{N_B}$

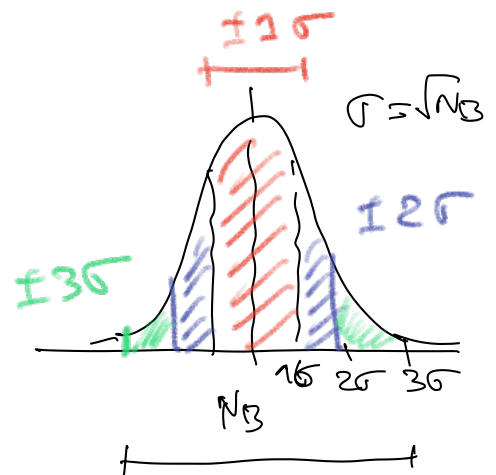
$$N_B = 10$$

$$N_{obs} = 13$$

$$= 7$$

$$\sqrt{N_B} \approx 3$$

$$\frac{N_{obs} - N_B}{\sqrt{N_B}} \approx 1$$



Compatible with N_B fluctuating to $N_B + \sqrt{N_B}$

$$N_{obs} = 21$$

$$\text{significance} = \frac{21 - 10}{3} \approx \frac{11}{3} \approx 4 \geq 3 \quad \text{evidence for signal}$$

$$N_{obs} = 30$$

$$\text{significance} = \frac{30 - 10}{3} \approx 7 > 5 \quad \text{Observation of signal}$$

Structure of nuclei: and proton

Rutherford scattering.

$\alpha + N \rightarrow \alpha + N$ elastic scattering.

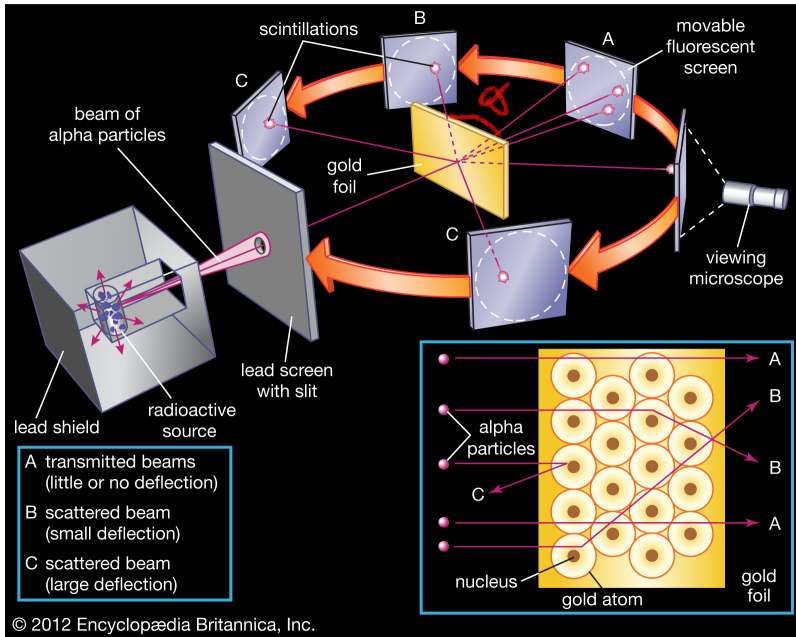
α : $M = 3.7 \text{ GeV}$

$K_\alpha = 5-10 \text{ MeV}$

$$E = m_\alpha + K \approx m_\alpha$$

$$\alpha \rightarrow \underline{P}_1 = (E, \vec{P}_1)$$

$$\begin{aligned} \alpha &\rightarrow \underline{P}_3 = (E', \vec{P}') \\ M &\rightarrow \underline{P}_4 = (E_4, \vec{P}_4') \\ &\quad \underline{P}_2 = (M, \vec{0}) \end{aligned}$$



$$\frac{dN}{dt} = \sigma \frac{dN}{dt} n_B \cdot d$$

↓
thickness

Differential cross section

$$\frac{d\sigma}{d\Omega} = \frac{d\sigma}{2\pi d\cos\theta}$$

$$d\Omega = d\phi d\cos\theta$$

$$\int_0^{2\pi} d\Omega = 2\pi d\cos\theta$$

$$\frac{d\sigma}{d\Omega} \approx \underbrace{\frac{(\alpha Z_1 Z_2)^2}{q^4}}_{\text{Rutherford}} \underbrace{\frac{|\vec{P}_{out}|}{|\vec{P}_{in}|}}_{\sim} E'^2 (1 - \beta^2 \sin^2 \frac{\theta}{2})$$

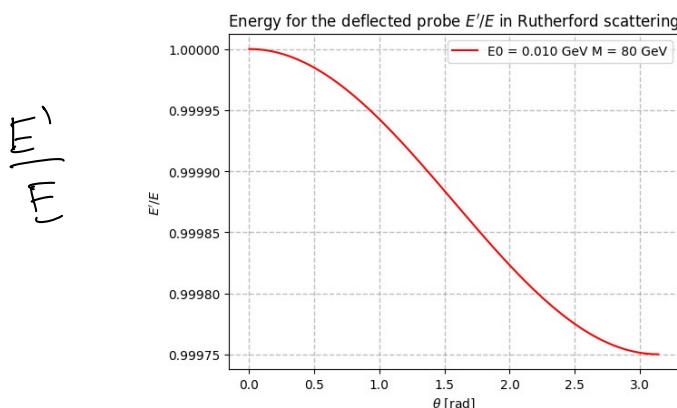
$$\begin{aligned} &Z_1 (E, \vec{P}_{in}) \rightarrow Z_2 (E', \vec{P}_{out}) \\ &\quad M, Z_2 \\ &\quad \beta = \frac{|\vec{P}_{in}|}{E} \end{aligned}$$

$$|\vec{P}_{out}| = |\vec{P}_{in}|$$

$$q = Z |\vec{P}_{in}| \sin \frac{\theta}{2}$$

$$E = 10 \text{ MeV}$$

$$M = 80 \text{ GeV}$$



$$\beta \approx 0$$

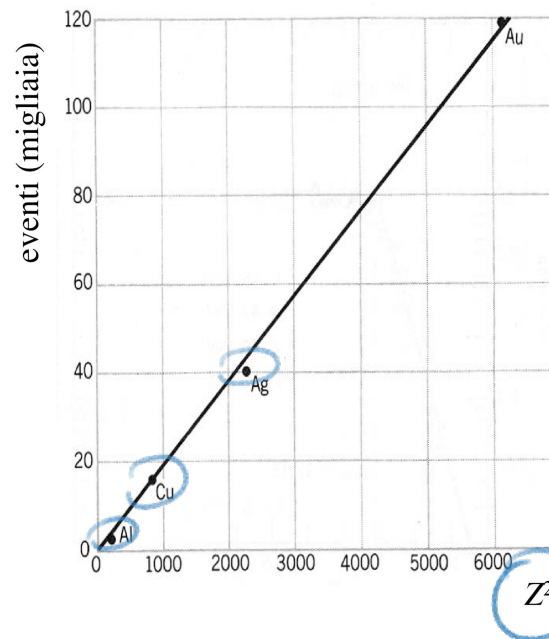
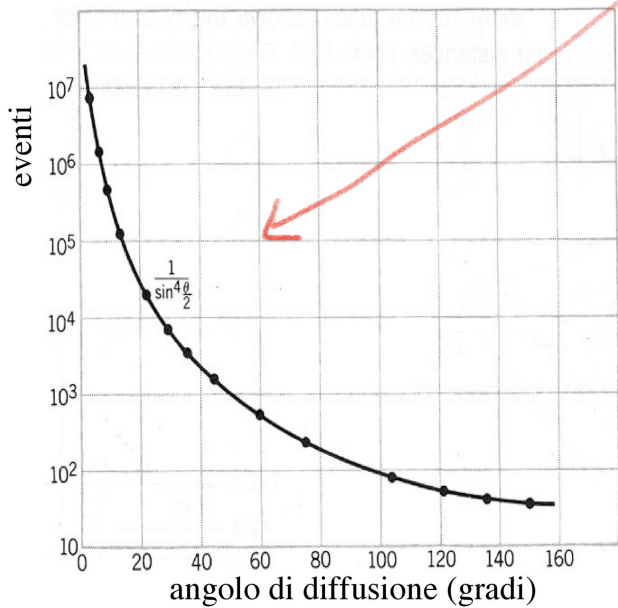
$$E_1^2 (1 - \beta^2 \sin^2 \frac{\theta}{2}) \approx m_1^2$$

$$\Rightarrow \frac{d\sigma}{d\Omega} \approx \frac{(\alpha Z_1 Z_2)^2}{16 P_{in}^4 \sin^4 \frac{\theta}{2}} m_1^2$$

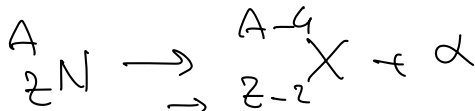
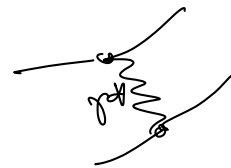
$$Z = Z(\alpha)$$

recoil is negligible

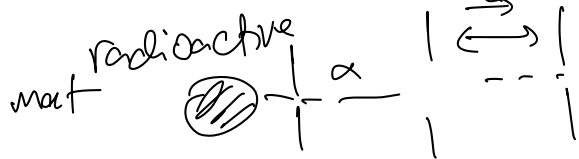
$$E' = E$$



$$\left(\frac{d\sigma}{d\Omega} \right)_{Ruth} \approx \frac{(\alpha Z_1 Z_2)^2}{\sin^4 \frac{\theta}{2}}$$



$$K\alpha \approx 5-10 \text{ MeV}$$



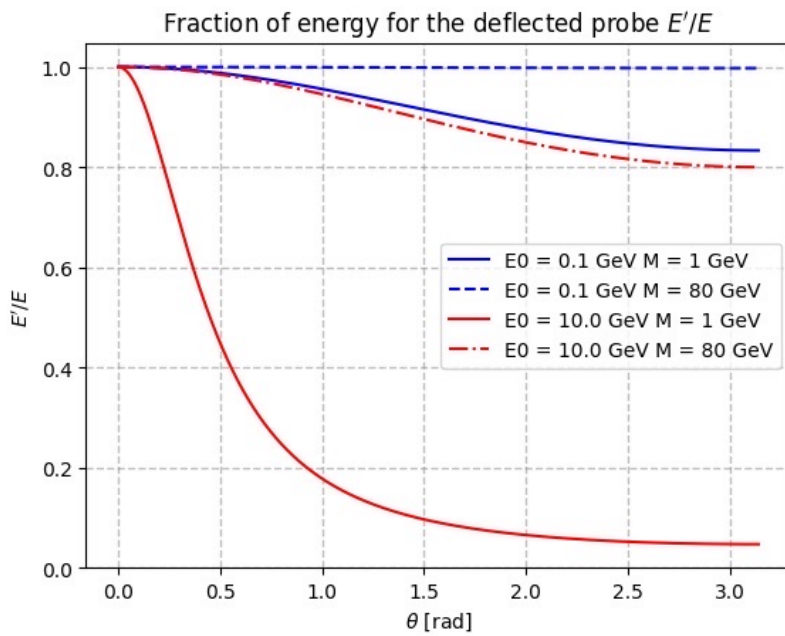
$$\Delta E = 90 \text{ V} = 2 \cdot e \Delta V$$

$$\Delta E = 20 \text{ MeV} \Rightarrow \Delta V = 10^7 \text{ V}$$

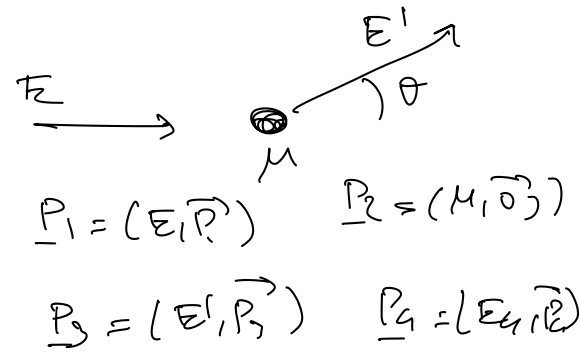
$$\frac{d\sigma}{d\Omega} \approx \underbrace{\frac{(\alpha Z_1 Z_2)^2}{q^4}}_{\text{Rutherford}} \underbrace{\frac{|\vec{P}_{out}|}{|\vec{P}_{in}|}}_{\text{Recoil}} \underbrace{E_1^2 (1 - \beta^2 \sin^2 \frac{\theta}{2})}_{\text{Spin}}$$

point like target

$$\frac{d\sigma}{d\Omega} \approx \underbrace{|M|^2}_{M_I} \underbrace{\rho(E)}_{\text{phase space.}}$$



$$\frac{E'}{E} = \frac{1}{1 + \frac{E}{M} (1 - \cos \theta)}$$



$$\underline{P}_1 + \underline{P}_2 = \underline{P}_3 + \underline{P}_4$$

$$e^- + p \rightarrow$$

$$M_p = 1 \text{ GeV}$$

$$E_e = 100 \text{ MeV}$$

Experiments: $\alpha + N \rightarrow \alpha + N$ with high energy α .

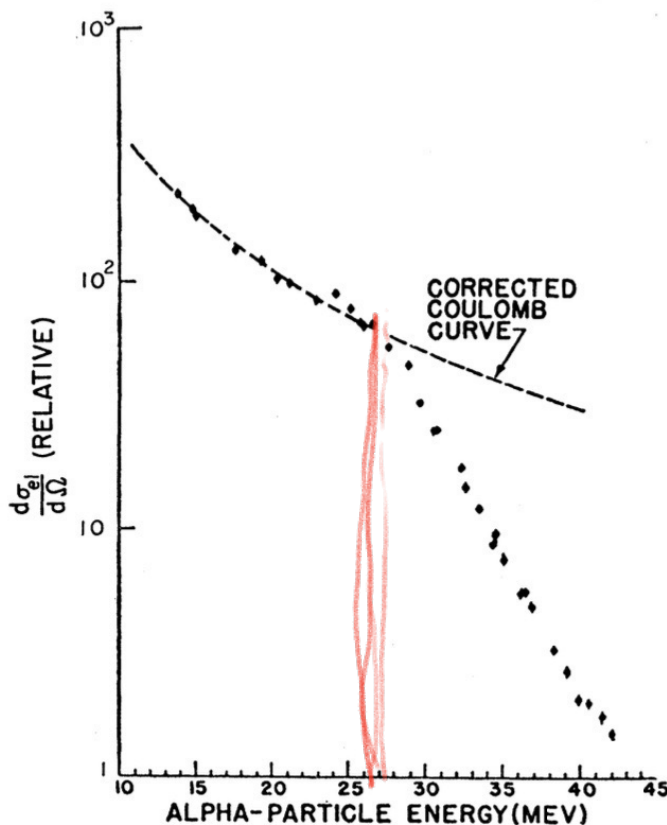


FIG. 5. Differential cross section for the elastic scattering of alpha particles by Pb at 60° as a function of the alpha-particle energy.

Rev. Mod. Phys. 33, 190 (1961)

$$\alpha + \text{Pb} \rightarrow \alpha + \text{Pb}$$

$$\text{Pb}: A \approx 208, Z = 82$$

$$\alpha: A = 4, Z = 2$$

Discrepancy between $\left. \frac{d\sigma}{d\Omega} \right|_{\text{Mott}} \neq \text{data}$.

$$E > 25 \text{ MeV}$$

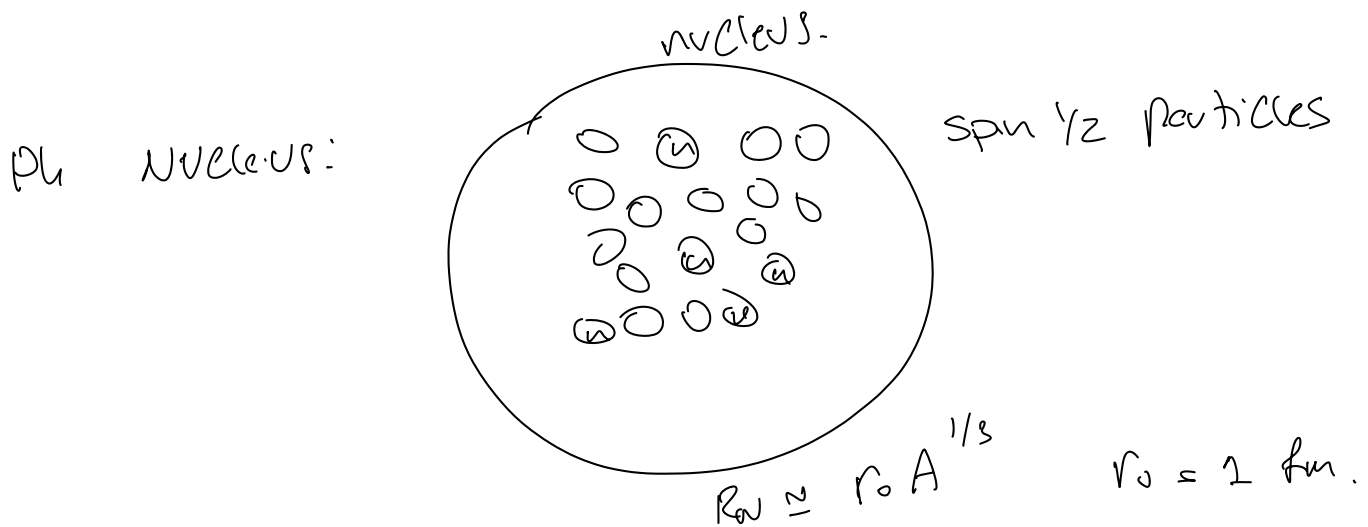
$$M_e = 0.5 \text{ MeV}$$

$$\frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega} \right)_{\text{Ruth.}} \frac{E'}{E} E'^2 (1 - \beta^2 \sin^2 \frac{\theta}{2})$$

$$\frac{1}{1 + \frac{E}{\mu} (1 - \cos \theta)}$$

$E = 25 \text{ MeV} \Rightarrow \beta \approx 1.$ electrons $e^- + N \rightarrow e^- + N.$

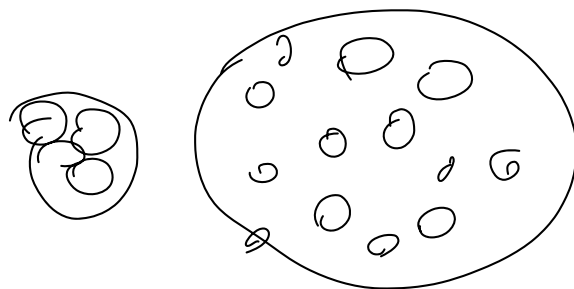
$$\frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega} \right)_{\text{Ruth.}} \frac{E'}{E} E'^2 \cos^2 \frac{\theta}{2}$$



neutrons, protons have Fermi energy inside nucleus.

$E_F \approx 20 \text{ MeV}.$ $K_F = \frac{p_F^2}{2m}$

$\alpha + N$ collision:



$E \geq 25 \text{ MeV} \Rightarrow$ better to use electrons.

$e^- \rightarrow$

$e^- + N \rightarrow e^- + N^{(*)}$

$E_e \geq 25 \text{ MeV}.$

$\beta \approx 1.$

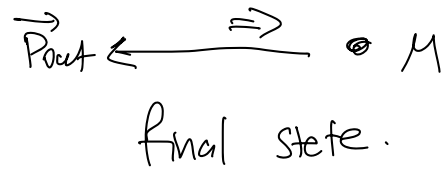
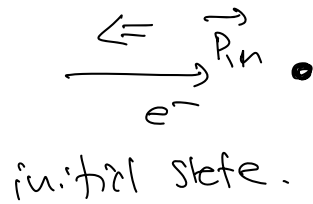
$$\frac{d\sigma}{d\Omega} = \left(\frac{\alpha Z_N}{q^4} \right)^2 E'^2 \cos^2 \frac{\theta}{2}$$

$$\theta \approx \pi \Rightarrow \frac{d\sigma}{d\Omega} \approx 0$$



helicity: $h = \frac{\vec{P} \cdot \vec{S}}{|\vec{P}| |\vec{S}|}$

chirality: $\psi = \underbrace{(1+\gamma_5)}_{P_L} \psi + \underbrace{(1-\gamma_5)}_{P_R} \psi = \psi_L + \psi_R$

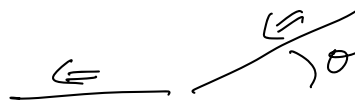


$$J = -1/2$$

$$+1/2$$

$$\Delta J = 1 \quad \text{not allowed.}$$

$$\Rightarrow \frac{d\sigma}{d\Omega} \propto \cos^2 \frac{\theta}{2}$$



$\Rightarrow \boxed{e^- + N \rightarrow e^- + N}$ elastic scattering.
with increasing energy.

$$\frac{d\sigma}{d\Omega} = \left(\frac{\alpha Z_N}{q^4} \right)^2 E^2 \cos^2 \frac{\theta}{2}$$

if this holds at all energies $E \rightarrow \infty$
 $\Rightarrow N$ is pointlike.

$$e^- + N \rightarrow e^- + X \quad \text{inelastic scattering}$$

Before inelastic scattering: excite nucleus.

$$e^- + N \rightarrow e^- + N^*$$

